

Vehicle Dynamics with Dugoff Tire Model: From Kinematic to Dynamic Simulation

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Abstract

This document presents the transition from a kinematic bicycle model to a dynamic vehicle model incorporating the Dugoff tire model. We derive the complete force-based dynamics, normal load distribution with load transfer, slip ratio computation, and friction-limited tire forces. This enables realistic simulation of traction limits, wheel slip, and combined longitudinal-lateral dynamics for autonomous mining haul truck applications.

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1 Introduction

1.1 Motivation for Dynamic Model

The kinematic bicycle model assumes:

- **No slip:** Pure rolling condition at all times
- **Unlimited traction:** Tire forces can be arbitrarily large
- **Instantaneous response:** No force buildup dynamics

These assumptions break down when:

- Vehicle operates on low-friction surfaces (gravel, wet haul roads)
- Emergency braking or aggressive acceleration
- Combined maneuvers (braking while turning)
- Realistic traction control and ABS simulation required

The Dugoff tire model addresses these by:

- Computing friction-limited forces based on normal load
- Modeling slip ratios (longitudinal and lateral)
- Capturing smooth saturation from linear to sliding regimes

1.2 Model Architecture Evolution

Kinematic Model (V1):

$$\text{Actuator} \rightarrow \text{Drive} \rightarrow \text{Velocity} \rightarrow \text{Position/Yaw} \quad (1)$$

Dynamic Model (V2):

$$\text{Actuator} \rightarrow \text{Drive} \rightarrow \boxed{\text{Tire Forces}} \rightarrow \text{Acceleration} \rightarrow \text{Velocity} \rightarrow \text{Position/Yaw} \quad (2)$$

where Tire Forces are computed by the Dugoff model based on slip and normal load.

2 Force-Based Longitudinal Dynamics

2.1 Equations of Motion

Newton's second law in the longitudinal direction:

$$m \frac{dv}{dt} = \sum F_x - F_{\text{drag}} - F_{\text{roll}} \quad (3)$$

where:

$$\sum F_x = F_{x,\text{fl}} + F_{x,\text{fr}} + F_{x,\text{rl}} + F_{x,\text{rr}} \quad (\text{sum of tire forces}) \quad (4)$$

$$F_{\text{drag}} = \frac{1}{2} \rho C_d A v^2 \quad (5)$$

$$F_{\text{roll}} = C_{\text{roll}} \quad (6)$$

2.2 Tire Force Allocation

For a rear-wheel-drive vehicle:

$$F_{x,\text{fl}} = 0 \quad (\text{front wheels: unpowered}) \quad (7)$$

$$F_{x,\text{fr}} = 0 \quad (8)$$

$$F_{x,\text{rl}} = f_{\text{Dugoff}}(\sigma_{x,\text{rl}}, \sigma_{y,\text{rl}}, F_{z,\text{rl}}) \quad (9)$$

$$F_{x,\text{rr}} = f_{\text{Dugoff}}(\sigma_{x,\text{rr}}, \sigma_{y,\text{rr}}, F_{z,\text{rr}}) \quad (10)$$

where f_{Dugoff} is the tire model function (detailed in Section 4).

2.3 Comparison: Kinematic vs Dynamic

Property	Kinematic	Dynamic
Traction limit	Unlimited	μF_z
Wheel slip	None	Computed
Braking distance	Underestimated	Realistic
Surface friction	Not modeled	$\mu \in [0.1, 0.9]$
Computational cost	Low	Medium

Table 1: Kinematic vs dynamic model comparison

3 Normal Load Distribution

3.1 Static Load Distribution

Weight distribution between front and rear axles:

$$F_{z,\text{front}}^{\text{static}} = \frac{mg \cdot d_{\text{rear}}}{L} \quad (11)$$

$$F_{z,\text{rear}}^{\text{static}} = \frac{mg \cdot d_{\text{front}}}{L} \quad (12)$$

where:

- m – vehicle mass (kg)
- $g = 9.81 \text{ m/s}^2$ – gravitational acceleration
- L – wheelbase (m)
- d_{front} – distance from CoG to front axle (m)
- d_{rear} – distance from CoG to rear axle (m)

Constraint: $d_{\text{front}} + d_{\text{rear}} = L$.

3.2 Longitudinal Load Transfer

During acceleration or braking, normal load shifts between axles:

$$\Delta F_z = \frac{ma_x h_{\text{CoG}}}{L} \quad (13)$$

where:

- a_x – longitudinal acceleration (m/s²), positive = acceleration
- h_{CoG} – center of gravity height (m)

Dynamic normal loads:

$$F_{z,\text{front}} = F_{z,\text{front}}^{\text{static}} - \Delta F_z \quad (14)$$

$$F_{z,\text{rear}} = F_{z,\text{rear}}^{\text{static}} + \Delta F_z \quad (15)$$

Physical interpretation:

- $a_x > 0$ (acceleration): Weight transfers to rear, $F_{z,\text{rear}}$ increases
- $a_x < 0$ (braking): Weight transfers to front, $F_{z,\text{front}}$ increases

3.3 Per-Wheel Normal Loads

Assuming equal left/right distribution (no lateral load transfer in V2):

$$F_{z,\text{fl}} = F_{z,\text{fr}} = \frac{F_{z,\text{front}}}{2} \quad (16)$$

$$F_{z,\text{rl}} = F_{z,\text{rr}} = \frac{F_{z,\text{rear}}}{2} \quad (17)$$

Note: Lateral load transfer during cornering is a future enhancement (V3).

3.4 Example: XCMG XDE320 Mining Truck

Parameters:

$$m = 218,000 \text{ kg (empty)} \quad (18)$$

$$L = 6.3 \text{ m} \quad (19)$$

$$h_{\text{CoG}} = 3.2 \text{ m} \quad (20)$$

$$d_{\text{front}} = 2.52 \text{ m (40\% of wheelbase)} \quad (21)$$

$$d_{\text{rear}} = 3.78 \text{ m (60\% of wheelbase)} \quad (22)$$

Static loads:

$$F_{z,\text{front}}^{\text{static}} = \frac{218,000 \times 9.81 \times 3.78}{6.3} = 1,288,000 \text{ N} \quad (23)$$

$$F_{z,\text{rear}}^{\text{static}} = \frac{218,000 \times 9.81 \times 2.52}{6.3} = 862,000 \text{ N} \quad (24)$$

Load transfer during braking at $a_x = -2.0 \text{ m/s}^2$:

$$\Delta F_z = \frac{218,000 \times (-2.0) \times 3.2}{6.3} = -222,000 \text{ N} \quad (25)$$

$$F_{z,\text{front}} = 1,288,000 - (-222,000) = 1,510,000 \text{ N} \quad (26)$$

$$F_{z,\text{rear}} = 862,000 + (-222,000) = 640,000 \text{ N} \quad (27)$$

Result: Front axle load increases by 17%, rear decreases by 26% during braking.

4 Dugoff Tire Model

4.1 Fundamental Equations

Tire forces are given by:

$$F_x = C_x \sigma_x f(\lambda) \quad (28)$$

$$F_y = C_y \sigma_y f(\lambda) \quad (29)$$

where:

- F_x, F_y – longitudinal and lateral tire forces (N)
- C_x, C_y – slip stiffness coefficients (N), load-dependent
- σ_x, σ_y – longitudinal and lateral slip ratios
- $f(\lambda)$ – friction saturation function
- λ – friction utilization parameter

4.2 Slip Ratios

Longitudinal slip ratio:

$$\sigma_x = \frac{\omega R - V_x}{V_x} \quad (30)$$

where:

- ω – wheel angular velocity (rad/s)
- R – tire effective radius (m)
- V_x – vehicle longitudinal velocity (m/s)

Sign convention:

$$\sigma_x > 0 \quad \text{Driving (wheel faster than vehicle)} \quad (31)$$

$$\sigma_x < 0 \quad \text{Braking (wheel slower than vehicle)} \quad (32)$$

$$\sigma_x = 0 \quad \text{Free rolling} \quad (33)$$

Lateral slip ratio:

$$\sigma_y = \frac{V_y}{V_x} \approx \tan(\alpha) \approx \alpha \quad (34)$$

where α is tire slip angle (rad), valid for small angles ($|\alpha| < 0.2$ rad).

4.3 Load-Dependent Slip Stiffness

Tire stiffness increases with normal load following an empirical power law:

$$C_x(F_z) = C_{x,\text{base}} \left(\frac{F_z}{F_{z,\text{ref}}} \right)^n \quad (35)$$

$$C_y(F_z) = C_{y,\text{base}} \left(\frac{F_z}{F_{z,\text{ref}}} \right)^n \quad (36)$$

where:

- $C_{x,\text{base}}, C_{y,\text{base}}$ – baseline stiffness at reference load (N)
- $F_{z,\text{ref}}$ – reference normal load (N), typically 800 kN for mining tires
- n – load exponent, typically 0.5 (square-root dependence)

Rationale: Contact patch area increases with load, leading to increased grip.

4.4 Friction Utilization Parameter

The key parameter governing saturation:

$$\lambda = \frac{\mu F_z}{2\sqrt{C_x^2\sigma_x^2 + C_y^2\sigma_y^2}} \quad (37)$$

where μ is the coefficient of friction (surface-dependent).

Physical interpretation:

$$\lambda \geq 1 \quad \text{Linear regime (no saturation)} \quad (38)$$

$$\lambda < 1 \quad \text{Nonlinear regime (friction-limited)} \quad (39)$$

4.5 Friction Saturation Function

$$f(\lambda) = \begin{cases} (2 - \lambda)\lambda = 2\lambda - \lambda^2, & \lambda < 1 \\ 1, & \lambda \geq 1 \end{cases} \quad (40)$$

Properties:

- Continuous at $\lambda = 1$: $\lim_{\lambda \rightarrow 1^-} f(\lambda) = 1$
- Smooth transition from linear to saturated regime
- Maximum value: $f_{\max} = 1$ at $\lambda = 1$

4.6 Proof: Force Saturation

Theorem: The total tire force never exceeds the friction limit:

$$\sqrt{F_x^2 + F_y^2} \leq \mu F_z \quad (41)$$

Proof: For $\lambda < 1$ (friction-limited case):

$$\sqrt{F_x^2 + F_y^2} = \sqrt{(C_x\sigma_x f)^2 + (C_y\sigma_y f)^2} \quad (42)$$

$$= f(\lambda) \sqrt{C_x^2\sigma_x^2 + C_y^2\sigma_y^2} \quad (43)$$

$$= (2\lambda - \lambda^2) \cdot \frac{\mu F_z}{2\lambda} \quad (\text{from definition of } \lambda) \quad (44)$$

$$= (2 - \lambda) \frac{\mu F_z}{2} \quad (45)$$

$$< \mu F_z \quad (\text{since } 0 < \lambda < 1) \quad (46)$$

For $\lambda \geq 1$, the tire is not friction-limited by construction. $\square$

5 Lateral Velocity Computation

5.1 Front Axle Lateral Velocity

Lateral velocity at front tire contact patch:

$$V_{y,\text{front}} = V_y + \dot{\psi} \cdot d_{\text{front}} + V_x \sin(\delta) \quad (47)$$

where:

- V_y – vehicle lateral velocity at CoG (m/s)
- $\dot{\psi}$ – yaw rate (rad/s)
- δ – steering angle (rad)

For small angles: $\sin(\delta) \approx \delta$.

5.2 Rear Axle Lateral Velocity

$$V_{y,\text{rear}} = V_y - \dot{\psi} \cdot d_{\text{rear}} \quad (48)$$

5.3 Yaw Rate from Bicycle Model

From kinematic bicycle model:

$$\dot{\psi} = \frac{V_x}{L} \tan(\delta) \approx \frac{V_x \delta}{L} \quad (49)$$

6 Implementation Algorithm

6.1 Computation Sequence

Step 1: Compute Normal Loads

1. Static distribution: $F_{z,\text{front}}^{\text{static}}, F_{z,\text{rear}}^{\text{static}}$
2. Load transfer: $\Delta F_z = \frac{ma_x h_{\text{CoG}}}{L}$
3. Dynamic loads: $F_{z,\text{front}} = F_{z,\text{front}}^{\text{static}} - \Delta F_z$
4. Per-wheel: $F_{z,\text{fl}} = F_{z,\text{fr}} = F_{z,\text{front}}/2$

Step 2: Compute Slip Ratios

1. Get wheel angular velocities: $\omega_{\text{fl}}, \omega_{\text{fr}}, \omega_{\text{rl}}, \omega_{\text{rr}}$
2. Longitudinal slip: $\sigma_x = (\omega R - V_x)/V_x$
3. Lateral velocity: $V_y = V_y(\dot{\psi}, \delta)$ from Section 5
4. Lateral slip: $\sigma_y = V_y/V_x$

Step 3: Compute Tire Forces (Dugoff)

1. Load-dependent stiffness: $C_x(F_z), C_y(F_z)$
2. Friction parameter: $\lambda = \mu F_z / (2\sqrt{C_x^2 \sigma_x^2 + C_y^2 \sigma_y^2})$
3. Saturation function: $f(\lambda)$
4. Forces: $F_x = C_x \sigma_x f(\lambda), F_y = C_y \sigma_y f(\lambda)$

Step 4: Update Dynamics

1. Total longitudinal force: $\sum F_x = F_{x,\text{rl}} + F_{x,\text{rr}}$
2. Net force: $F_{\text{net}} = \sum F_x - F_{\text{drag}} - F_{\text{roll}}$
3. Acceleration: $a = F_{\text{net}}/m$
4. Velocity: $v_{k+1} = v_k + a\Delta t$

6.2 Discrete-Time Update

Complete state update for timestep Δt :

$$a_k = \frac{F_{x,rl} + F_{x,rr} - F_{\text{drag}} - F_{\text{roll}}}{m} \quad (50)$$

$$v_{k+1} = v_k + \Delta t \cdot a_k \quad (51)$$

$$\dot{\psi}_k = \frac{v_k}{L} \tan(\delta_k) \quad (52)$$

$$\psi_{k+1} = \psi_k + \Delta t \cdot \dot{\psi}_k \quad (53)$$

$$x_{k+1} = x_k + \Delta t \cdot v_k \cos(\psi_k) \quad (54)$$

$$y_{k+1} = y_k + \Delta t \cdot v_k \sin(\psi_k) \quad (55)$$

7 Surface Friction Coefficients

7.1 Mining Haul Road Conditions

Surface	μ_{peak}	μ_{slide}
Dry pavement (smooth)	0.85	0.75
Gravel compact (baseline)	0.72	0.65
Gravel loose	0.55	0.48
Iron ore dust (dry)	0.45	0.38
Iron ore dust (wet)	0.30	0.25
Mud	0.25	0.20
Sand (loose)	0.40	0.35

Table 2: Friction coefficients for Pilbara mining conditions

7.2 Velocity-Dependent Friction (Optional)

Friction reduces at higher speeds due to reduced tire-road contact time:

$$\mu_{\text{effective}}(V) = \mu_{\text{peak}} \cdot (1 - k_v |V|) \quad (56)$$

where $k_v = 0.003 \text{ s/m}$ (typical fade factor).

Constraint: $\mu_{\text{effective}} \geq 0.70\mu_{\text{peak}}$ (minimum 70% retention).

8 Validation and Benchmarks

8.1 Expected Behavior

Scenario 1: Acceleration on gravel ($\mu = 0.72$)

XCMG XDE320 loaded (538 tons):

$$F_{z,\text{rear}} \approx 0.75 \times 538,000 \times 9.81 = 3,956,000 \text{ N} \quad (57)$$

$$F_{x,\text{max}} = 0.72 \times 3,956,000 = 2,848,000 \text{ N} \quad (58)$$

$$F_{x,\text{motor}} = \frac{145,000}{1.93} = 75,100 \text{ N} \quad (59)$$

Result: Motor force $\ll$ traction limit $\rightarrow$ power-limited, not traction-limited.

Scenario 2: Emergency braking on wet dust ($\mu = 0.30$)

$$F_{x,\max} = 0.30 \times 538,000 \times 9.81 = 1,585,000 \text{ N} \quad (60)$$

$$F_{x,\text{brake}} = \frac{180,000}{1.93} = 93,300 \text{ N} \quad (61)$$

Result: Still power-limited, but much closer to traction limit.

8.2 Stopping Distance Comparison

Kinematic model: Assumes perfect traction, underestimates distance.

Dynamic model: Friction-limited, realistic distance:

$$d_{\text{stop}} = \frac{v_0^2}{2\mu g} \quad (62)$$

For $v_0 = 64 \text{ km/h} = 17.78 \text{ m/s}$ on gravel ($\mu = 0.72$):

$$d_{\text{stop}} = \frac{17.78^2}{2 \times 0.72 \times 9.81} = 22.4 \text{ m} \quad (63)$$

9 Future Enhancements

9.1 Short-Term (V2.5)

- Lateral load transfer during cornering
- Combined longitudinal-lateral friction circle
- Wheel dynamics (rotational inertia)

9.2 Medium-Term (V3)

- Pacejka "Magic Formula" tire model (higher fidelity)
- Suspension dynamics (vertical motion, damping)
- Thermal tire effects

9.3 Long-Term (V4)

- 3D terrain interaction (grades, banking)
- Tire wear modeling
- Multi-physics coupling (thermal-mechanical)

10 Conclusion

The Dugoff tire model provides a significant improvement over the kinematic bicycle model by:

- Enforcing realistic traction limits based on surface friction
- Computing slip ratios and forces explicitly
- Enabling ABS, traction control, and stability control validation
- Supporting realistic braking distance and emergency maneuver simulation

This forms the foundation for dynamic vehicle simulation suitable for autonomous mining vehicle development.

References

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